AUDIT SURFACE FOR

Capability Matters: A Casebook

Validation & Audit

INDEXES · PER-CASE REFERENCES

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SECTION

What this document is for

Validation and Audit is the auditor-facing companion to **Capability Matters: A Casebook**. It carries the surfaces an instructor, reviewer, or accreditation reader needs to verify the casebook — the indexes that let any case be located by domain or by LENS course, and the per-case reference appendix that lets any source be traced. It is the same data the casebook itself ships, lifted into a standalone document so the LENS Companion can stay focused on what LENS **is** (the five competencies, the LEOs, the course mapping) without the audit pages travelling with it.

The document is in three parts:

I. Cases by primary domain — every case classified by the operational domain it primarily evidences, so a domain expert can find the cases that touch their territory without scanning the table of contents.

II. Cases by LENS course — every case classified by the LENS course it serves as a worked example for, so syllabus authors and capstone advisors can find the cases that fit a course's stated learning outcomes.

III. References, case by case — every reference cited in every case, organised by case in number order, with an explicit **Retrieved from:** line per entry so a reviewer can locate the canonical source.

> **On verification.** The running per-case verification track lives in the repository at `casebook/verification-log.md`, a 194-row table whose six auto-prefilled columns track the structural and sourcing checks the build pipeline can answer on its own. The seventh column — **conclusions reasonable** — is the human case-by-case content read this document supports. The references appendix below is the per-case checklist for that read; the **Retrieved from:** lines are the canonical-access record each case acquires as the verification pass closes.

DOMAIN INDEX

The cases by domain.

The matrix at the front lists every case in number order. This index reorganizes the cases by primary domain — a reader following healthcare, defense, education, or any other thread can find the relevant cases together. Each case appears once, under its primary domain. Where a section opens with a callout, it names the dataset's standout failure and, where one exists in that domain, its standout success.

DEFENSE

STANDOUT FAILURE — **Case 128** — **Operation Eagle Claw (1980).** Cross-organizational capability that existed on no service's deliverable list; the mission was improvised across the Army, Navy, Marines, Air Force, and CIA. Produced the Holloway Commission, USSOCOM, and Goldwater-Nichols.

STANDOUT SUCCESS — **Case 137** — **U.S. Nuclear Navy / Rickover Training Model (1954–).** The longest continuous capability-engineering program in any high-consequence domain. Sixty-plus years of zero reactor accidents through training treated as a system parameter.

124	USS Fitzgerald & USS John S. McCain	2017	T·K·N
126	F-35 Sustainment & the Maintainer Capability Gap	ongoing	T·K·D
127	Military Fratricide — Desert Storm to Afghanistan	1991 – 2014	T·H·K
128	Operation Eagle Claw	1980	T·K
129	Patriot Missile / Dhahran	1991	D·H·K
130	USS Vincennes & Iran Air Flight 655	1988	H·T
131	Stanislav Petrov / 1983 False Alert	1983	H·T
133	Mark 14 Torpedo Failures	1941 – 1943	T·K·G
134	V-22 Osprey	1991 – present	T·H·N
135	9/11 Intelligence Sharing Failures	1996 – 2001	G·K
137	U.S. Nuclear Navy / Rickover Training Model	1954 – present	T·K·N
138	MIL-STD-1472H — Human Engineering as a Binding Acquisition Standard	2020 revision (series since 1968)	G·K

AVIATION

STANDOUT FAILURE **Case 97** **Boeing 737 MAX / MCAS (2018–19).** A single-string angle-of- attack sensor, a flight-control law that assumed pilot mental models from prior models, and a certification chain that did not knit the pieces together. 346 lives.

STANDOUT SUCCESS **Case 117** **Crew Resource Management & CAST (1981–).** Authority gradients named and engineered into the cockpit, then a closed-loop industry data architecture sixteen years later. The 83% fatality-risk reduction CAST reports is a portfolio result; CRM's separate contribution is not isolated in any published estimate.

132	F-22 OBOGS Hypoxia — The Sustainment-Era Instrumentation Gap	2008 – 2012	H·K·N
185	Air Canada Chatbot Liability — Delegation Without Revocation	2022 – 2024	D·K·N

HEALTHCARE

STANDOUT FAILURE — **Case 2** — **EHR / CPOE Implementation (2005–).** Roughly $30B federal investment in systems designed for billing and administration, deployed against clinical workflow. Usability is now itself a patient-safety variable at scale.

STANDOUT SUCCESS — **Case 23** — **WHO Surgical Safety Checklist (2008–).** One page, nineteen items, paired with a required pause. Halved surgical mortality in the multi-site pilot; the smallest effective capability intervention in the dataset.

1	Therac-25	1985 – 1987	H·D·G
2	EHR / CPOE Implementation	2005 – present	H·D·G
4	Mid Staffordshire NHS Foundation Trust	2005 – 2009	G·N·K
5	Epic Sepsis Model	2017 – 2021	D·G·N
8	Medical Errors as Systemic Failure	1999 – present	T·H·N·K·G
9	Vioxx Withdrawal	1999 – 2004	G·D
11	California Nurse Staffing Ratios — Legislating a Capability Requirement	2004 – 2010	G·K
12	ACGME 80-Hour Resident Duty-Hour Reform	2003–2017	T·K·N
13	Implementation Science in Healthcare — The 17-Year Gap	ongoing	K·G·N
19	Keystone ICU / Pronovost Checklist	2004 – present	T·N
20	TREWS / Sepsis Watch	2018 – 2022	H·K·G
21	Sterile-Cockpit Ward Rounds — Adapting an Aviation Principle to Clinical Handoff	2024 – 2025	H·K·N
22	ChatGPT in Healthcare — Hallucination Cases	2023 – present	H·D
23	WHO Surgical Safety Checklist	2008 – present	T·N
24	Bristol Heart Babies Reform	1984 – present	G·N
32	Annual-Screening UI Redesign + CDS at University of Missouri Health Care	2020	T·D·N
33	Alert-Fatigue Redesign — Cutting EHR Alerts Without Cutting the Safety Signal	2019 – 2024	T·D·N
34	COMPOSER Sepsis Prediction — The Third Clinical-AI Sepsis Case	2022 – 2024	T·K·D
35	Radiology AI Miscalibration	2018 – present	H·K·D

36 AlphaFold — Protein Structure Prediction 2020 – present T

37 DeepMind Mammography — High-Profile Nature Paper, Replicability Pushback 2020 T·K·D

39 TeamSTEPPS 2006 – present T·N

176 Tylenol Recall 1982 G·N

EDUCATION AT SCALE

STANDOUT FAILURE — Case 54

Summit Learning / Personalized Learning Rollout (2014–19). A defensible pedagogical platform, well funded, deployed across 380 schools — and pulled by parent revolt because the governance architecture (consent, evidence, exit) was never engineered alongside it.

STANDOUT SUCCESS — Case 80

Georgia State University Predictive Analytics (2012–). 800 risk factors per student, advising triggered by early warning signs, equity as a primary constraint. Six-year graduation 32% → 54%; equity gaps eliminated.

45 Tennessee Voluntary Pre-K Study 2009–2018 G·D

46 Algorithmic Bias in Educational Predictive Analytics ongoing G·H·D

51 Atlanta Public Schools Cheating Scandal 2009 – 2015 G·N

53 inBloom 2014 G

54 Summit Learning / Personalized Learning Rollout 2014–2019 G·T·K

67 Cognitive Tutor / Carnegie Learning 1990s – present T

80 Georgia State University Predictive Analytics 2012 – present T·K

142 xAPI / Total Learning Architecture — Interoperability Gap ongoing K·G

205 The Discipline We Build Next ongoing T·K·N·G·H·D

ENERGY

STANDOUT FAILURE — Case 166

Three Mile Island (1979). Training matched the design-basis events the engineers had imagined; the accident was not one of them. The control-room interface confused command signal with valve state. Catalysed industry-wide reform.

STANDOUT SUCCESS — Case 175

INPO and the Nuclear Academy (1979–). The industry built the institution TMI revealed it needed. No INES Level 4-or-above accident at a U.S. commercial reactor since.

146 Northeast Blackout 2003 H·K

160	Davis-Besse Reactor Head Corrosion	2002	N·K·G
161	Texas City BP Refinery Explosion	2005	N·T·K·G
162	Upper Big Branch Mine Explosion	2010	N·G·K
163	Deepwater Horizon	2010	T·N·K
165	Camp Fire / PG&E	2018	G·N·K
166	Three Mile Island	1979	T·H
169	Fukushima Daiichi	2011	N·G·K
175	INPO and the Nuclear Academy	1979 – present	T·K·G

GOVERNMENT

STANDOUT FAILURE — Case 191

Australia Robodebt (2016–2023). An income-averaging algorithm that reversed the burden of proof onto welfare recipients, raising some 470,000 wrongful debts. The Royal Commission found the scheme unlawful and attributed deaths to its operation — circumstantially, not as a finding of individual causation.

NOTE — Case 203

The reforms that follow these failures — Royal Commissions, statutory restructures — are documented inside the narratives rather than as separate cases. The government dataset's paired-intervention success is NYC Local Law 144 (Case 203), filed under its own primary domain.

7	VA Wait-Time Scandal	2014	G·K·N
180	Healthcare.gov Launch	2013	K·T·G
193	Bernard Madoff / SEC Failures	1992 – 2008	G·K·N
197	Predictive Policing — PredPol	2011 – present	G·H·D

INDUSTRIAL

STANDOUT FAILURE — Case 159

Bhopal (1984). Training, knowledge, normalization of deviance, and governance all collapsed in the same plant. Thousands died within hours; total-death estimates run to 15,000–20,000, with roughly half a million exposed. The cautionary case for outsourced operational responsibility.

STANDOUT SUCCESS — Case 155

Toyota Production System / Andon Cord (1950s–). Any operator can signal a defect at the source. Under the fixed-position stop system most pulls never halt the line — the smallest authority-architecture intervention in the dataset, and the most durably copied.

144	Kodak Digital Camera Stagnation	1975–2012	D·K·N

TECHNOLOGY

STANDOUT FAILURE — **Case 184** — **UK Post Office Horizon Scandal (1999–2015).** Institutional refusal to believe sub-postmasters' reports of computer error, sustained over fifteen years. More than 900 wrongful convictions, 236 imprisoned, and 13 suicides the inquiry could not rule out as Horizon-caused — the longest-running operator-feedback failure in the dataset.

NOTE — **Case 36** — The technology dataset's success story is filed under Healthcare: AlphaFold (Case 36), the contemporary worked example of computational capability-engineering done as a discipline.

AI IN EDUCATION

K-12 EDUCATION

K-12 MATHEMATICS

K-12 SCIENCE

AIR TRAFFIC MANAGEMENT

ANESTHESIOLOGY

AUTONOMOUS SYSTEMS

STANDOUT FAILURE — Case 183

Uber ATG / Tempe Fatality (2018). The safety driver was nominally responsible for what the system was designed to do without her attention — the classic vigilance contradiction at the heart of partial-autonomy operator roles.

NOTE — Case 205

This key's cases are all failures and frontiers; the paired-intervention successes — Waymo's safety case (199) and the CPUC permit regime (200) — are filed under autonomous vehicles. See Case 205 (**The Discipline We Build Next**) for the open question this volume closes on.

AUTONOMOUS VEHICLES

AVIATION INFRASTRUCTURE

AVIATION SAFETY

CLINICAL CARE

CLINICAL MEDICINE

CLINICAL/TRANSLATIONAL RESEARCH

CORPORATE L&D

CRIMINAL JUSTICE

DIGITAL IDENTITY

E-GOVERNMENT

GOVERNMENT/WELFARE (NETHERLANDS)

HEALTH PROFESSIONS EDUCATION

HEALTHCARE/ONCOLOGY

HIGHER EDUCATION

HIGHER EDUCATION (AFRICA)

HIGHER EDUCATION (BRAZIL)

HIGHER EDUCATION (LATIN AMERICA)

HIGHER ED (UK)

HIGHER-ED ANALYTICS

HOSPITAL PHARMACY

HUMANITARIAN RELIEF

IMPLEMENTATION SCIENCE

LEARNING ANALYTICS

MEDICAL DEVICES

MEDICAL EDUCATION

MEDICAL-DEVICE REGULATION

MULTIMODAL LEARNING ANALYTICS

NAVAL ENGINEERING

NEUROSCIENCE

NEUROSCIENCE/CONNECTOMICS

NUCLEAR ENGINEERING

156 INL / LWRS Control-Room Modernization — Sustainment Research for an Aging Fleet 2010 – present D·H·K

NUCLEAR POWER

154 INL Turbine-Control Upgrade — Verification and Validation as the Cutover Deliverable 2014 G·K·H

NURSING EDUCATION

16 NCSBN National Simulation Study — Licensing the 50% Substitution Rule 2014 G·K·D

PHARMA SAFETY

38 iPLEDGE Isotretinoin REMS — When the Authorization Mechanism Underperforms 2006 – 2011 G·D·N

PUBLIC HEALTH

179 The Johns Hopkins COVID-19 Dashboard — Evidence Infrastructure at Decision Speed 2020 – 2023 G·K

RAIL TRANSPORT

177 CIRAS — Confidential Incident Reporting for UK Rail 1996 – present G·K·N

SOFTWARE ENGINEERING EDUCATION

71 Reflective Practice on a Work-Based Software Engineering Program — Longitudinal Capability Development 2025 (preprint) K·N

SOFTWARE SUSTAINMENT

174 Y2K Remediation — The Aging-System Transition That Worked Because It Was Believed 1996 – 2000 G·D·K

SPACE / AEROSPACE

STANDOUT FAILURE — Case 111 — **Challenger & Columbia (1986 / 2003).** Normalisation of deviance across seventeen years and two crews — the sociologist Diane Vaughan's term for a culture that accepted flying with flaws once problems were defined as normal and routine, and the culture the CAIB found still intact in 2003.

NOTE — Case 137 — No paired-intervention success in the space dataset — Part III's aviation half carries seven. See Defense (Rickover, Case 137) for the safety-engineering counterpart human spaceflight did not build.

98 Mars Climate Orbiter — Unit Mismatch 1999 D·K

101 Ariane 5 Flight 501 1996 D·K·H

About this index. Each case is filed under its primary domain, so every case appears exactly once. Cases that span several domains (a healthcare-AI case is also a technology case, a defense-training case is also an education case) are cross-referenced in their narratives. The standout failures and successes are editorial picks — the case in that domain the editors return to most often in studio.

LENS COURSE INDEX

The cases by LENS course.

Each case names the LEN courses it serves as a worked example for. This index inverts that mapping: for every course in the concentration, the cases that inform it, in case-number order. A case appears under every course it supports, so many appear more than once.

LEN 1

36 cases

LEN 2

53 cases

LEN 3

47 cases

LEN 4

68 cases

LEN 5

84 cases

LEN 6

17 cases

LEN 7

102 cases

LEN 8

90 cases

LEN 9

30 cases

LEN 10

21 cases

APPENDIX · REFERENCES BY CASE

References, case by case.

Every reference cited in every case, organized by case in number order. The intent is that any reader can locate every source the book draws on. Each entry preserves the in-case citation form (author, year, venue) and adds a **Retrieved from:** rule beneath it — a worksheet line for the reviewer to enter the canonical access path they used: a publisher URL, a DOI, an agency landing page, or a stable archival locator. The rules print blank by design, because the per-case verification is being carried out against this appendix rather than reported by it; the running status of that work is in `casebook/verification-log.md`. Future editions will print the locators as the review closes them out.

> **Style.** References are rendered in the form they appeared in the case — APA-style author/year/title/venue, with italicised book titles and journal names — supplemented by the explicit retrieval locator. Government and agency reports are cited by report number when one exists. Trade-book references that carry no DOI use ISBN as the stable identifier; that mapping is on the verification log.

CASE 1 Therac-25 1985 – 1987

1. N. G. Leveson & C. S. Turner, “An Investigation of the Therac-25 Accidents,” IEEE Computer 26(7): 18–41 (1993). doi:10.1109/MC.1993.274940 — the removed hardware interlocks and the adapted software. — Retrieved from: ____________________

2. Leveson & Turner (1993) — the two software defects (data-entry race condition and counter overflow), the uninformative “Malfunction 54”, overdoses up to 100×, six accidents, and at least three deaths. — Retrieved from: ____________________

3. Leveson & Turner (1993); N. G. Leveson, Safeware: System Safety and Computers (Addison-Wesley, 1995) — the manufacturer’s denial and the systemic findings. — Retrieved from: ____________________

4. N. G. Leveson, Engineering a Safer World: Systems Thinking Applied to Safety (MIT Press, 2011) — why removing a safeguard requires explicitly reassigning its safety function. — Retrieved from: ____________________

5. The case’s role in medical-device software regulation and software-safety practice (FDA software guidance; IEC 62304 lineage); see also Ethics Unwrapped, UT Austin. — Retrieved from: ____________________

CASE 2 EHR / CPOE Implementation 2005 – present

1. Health Information Technology for Economic and Clinical Health (HITECH) Act (2009) — the $30B EHR incentive program. — Retrieved from: ____________________

2. Y. Han et al., “Unexpected Increased Mortality After Implementation of a Commercially Sold CPOE System,” Pediatrics 116(6): 1506–1512 (2005) — the disputed pediatric-ICU mortality result. — Retrieved from: ____________________

3. J. Howe, K. Adams, A. Z. Hettinger & R. Ratwani, “Electronic Health Record Usability Issues and Potential Contribution to Patient Harm,” JAMA 319(12): 1276–1278 (2018) — 557 of 1.735M safety reports tied usability to possible harm. — Retrieved from: ____________________

4. D. Sittig & H. Singh, "Defining Health Information Technology-related Errors: New Developments Since To Err is Human," Archives of Internal Medicine 171(14): 1281–1284 (2011). — Retrieved from: ____________________

5. M. Grissinger, "The Absence of a Drug–Disease Interaction Alert Leads to a Child's Death," P&T 43(2): 71–72 (2018); F. Schulte & E. Fry, "Death by 1,000 Clicks," Fortune/KHN (2019). — Retrieved from: ____________________

6. Pew / MedStar / AMA, Ways to Improve Electronic Health Record Safety (2018); R. Wachter, The Digital Doctor (2015). — Retrieved from: ____________________

CASE 3 Watson for Oncology — Delegation Marketed Ahead of Capability 2013 – 2018

1. Ross & Swetlitz, "IBM's Watson supercomputer recommended 'unsafe and incorrect' cancer treatments, internal documents show," STAT (25 July 2018), and "IBM pitched its Watson supercomputer as a revolution in cancer care. It's nowhere close," STAT (5 Sept 2017) — the primary sources; investigative journalism drawing on leaked IBM internal documents. — Retrieved from: ____________________

2. Strickland (2019), "How IBM Watson Overpromised and Underdelivered on AI Health Care," IEEE Spectrum — independent retrospective synthesis of the public record. — Retrieved from: ____________________

3. Topol (2019), Deep Medicine, Basic Books — secondary academic situating of Watson within the broader clinical-AI delegation discourse. — Retrieved from: ____________________

4. Somashekhar et al. and the WFO lung/colorectal early-experience abstracts, J Clin Oncol 35 (2017), 15_suppl, presented at the ASCO Annual Meeting — the marketed concordance figures: 96% lung, 93% rectal, 81% colon. — Retrieved from: ____________________

5. v2 paired cases: TREWS (Case 20), Epic Sepsis Model (Case 5), SyRI (Case 189) — the AI-delegation typology. — Retrieved from: ____________________

CASE 4 Mid Staffordshire NHS Foundation Trust 2005 – 2009

1. R. Francis QC, Report of the Mid Staffordshire NHS Foundation Trust Public Inquiry (2013) — the staffing cuts and Foundation Trust targets. — Retrieved from: ____________________

2. Healthcare Commission, Investigation into Mid Staffordshire NHS Foundation Trust (2009) — ward conditions and excess mortality. — Retrieved from: ____________________

3. Francis QC (2013) — the three-volume, 1,782-page report and 290 recommendations. — Retrieved from: ____________________

4. Francis QC (2013) — "not just the Trust's Board but the system as a whole failed in its most essential duty — to protect patients from unacceptable risks of harm and from unacceptable, and in some cases inhumane, treatment" (quoted). — Retrieved from: ____________________

5. D. Berwick, A Promise to Learn — A Commitment to Act (National Advisory Group on the Safety of Patients in England, 2013). — Retrieved from: ____________________

6. K. Walshe & J. Higgins (2002) on NHS safety governance; cf. the VA wait-time scandal (Case 7). — Retrieved from: ____________________

CASE 5 Epic Sepsis Model 2017 – 2021

1. Wong et al. (2021), "External Validation of a Widely Implemented Proprietary Sepsis Prediction Model in Hospitalized Patients," JAMA Internal Medicine 181(8):1065–1070, doi:10.1001/jamainternmed.2021.2626. — Retrieved from: ____________________

2. Habib et al. (2021), commentary on Wong et al., JAMA Internal Medicine — on the implications for proprietary clinical AI. — Retrieved from: ____________________

3. FDA, Clinical Decision Support Software: Final Guidance (2022) — the post-Wong reframing of the EHR-embedded oversight question. — Retrieved from: ____________________

4. Ross, C. (2022), "Epic overhauls its sepsis algorithm," STAT News (Oct 2022) — the ESM v2 redesign and the shift to locally-trained models. — Retrieved from: ____________________

5. Adams et al. (2022), Nature Medicine — the paired positive case (20). — Retrieved from: ____________________

CASE 6 Racial Bias in Pain Assessment — The False-Belief Mechanism 2016

1. Hoffman, Trawalter, Axt, & Oliver (2016), "Racial bias in pain assessment and treatment recommendations, and false beliefs about biological differences between blacks and whites," PNAS 113(16):4296–4301, doi:10.1073/pnas.1516047113. — Retrieved from: ____________________

2. Anderson, Green, & Payne (2009), "Racial and ethnic disparities in pain: causes and consequences of unequal care," Journal of Pain 10(12):1187–1204 — the population-level disparity. — Retrieved from: ____________________

3. Sabin & Greenwald (2012), "The influence of implicit bias on treatment recommendations for 4 common pediatric conditions," American Journal of Public Health — the diffuse-mechanism backdrop the Hoffman precision improves on. — Retrieved from: ____________________

4. Vyas, Eisenstein, & Jones (2020), NEJM — connecting race-in-clinical-algorithms to race-in-clinical-judgment. — Retrieved from: ____________________

5. Omiye, Lester, Spichak, Rotemberg, & Daneshjou (2023), "Large language models propagate race-based medicine," npj Digital Medicine 6:195 — commercial LLMs reproduce the Hoffman false beliefs. — Retrieved from: ____________________

CASE 7 VA Wait-Time Scandal 2014

1. VA Office of Inspector General, Report 14-02603-267 (2014) — secret waiting lists and falsified appointment data. — Retrieved from: ____________________

2. GAO Veterans Health Administration reports (2000–2019) — fifteen years of data-reliability warnings. — Retrieved from: ____________________

3. VA OIG reports (2005, 2007, 2008) — prior, unactioned findings. — Retrieved from: ____________________

4. D. Draper (GAO), House VA Committee testimony, GAO-19-687T (2019) — "given the high turnover among schedulers, it is important that VA remain vigilant about scheduler training" (quoted). — Retrieved from: ____________________

5. Veterans Access, Choice, and Accountability Act (2014) — the legislative response. — Retrieved from: ____________________

6. C. Argyris & D. Schön, Organizational Learning: A Theory of Action Perspective (1978); G. Bevan & C. Hood, "What's Measured Is What Matters," Public Administration 84(3): 517–538 (2006). — Retrieved from: ____________________

CASE 8 Medical Errors as Systemic Failure 1999 – present

1. Institute of Medicine, To Err Is Human: Building a Safer Health System (1999); Crossing the Quality Chasm (2001); Improving Diagnosis in Health Care (2015) — the field-defining trilogy and the 44,000–98,000 estimate; the systems framing. — Retrieved from: ____________________

2. Makary, M. & Daniel, M. (2016), "Medical error — the third leading cause of death in the U.S.," BMJ 353:i2139 — the 250,000 estimate, the quoted framing, the extrapolation from four studies published since the IOM report (Health-Grades 2004; OIG 2010; Landrigan 2010; Classen 2011). — Retrieved from: ____________________

3. Shojania, K. & Dixon-Woods, M. (2017), "Estimating deaths due to medical error: the ongoing controversy and why it matters," BMJ Quality & Safety 26(5):423–428; with Bates, D. W. & Singh, H. (2018), Health Affairs 37(11):1736–1743 — methodological contestation of the Makary extrapolation and CDC-ranking comparison. — Retrieved from: ____________________

4. Makary & Daniel (2016), BMJ — death certificates do not capture medical error as a cause; ICD billing taxonomy as the structural reason. — Retrieved from: ____________________

5. Bates, D. W., Levine, D. M., Salmasian, H., et al. (2023), “The Safety of Inpatient Health Care,” NEJM 388(2):142–153 — eleven-hospital Massachusetts cohort of 2,809 admissions; at least one adverse event in 23.6% of admissions, 25% of those preventable; persistence of harm at scale. — Retrieved from: ____________________

6. Agency for Healthcare Research and Quality, National Healthcare Quality and Disparities Reports (annual); CDC WONDER ICD-coded mortality data — institutional context for the missing national active-surveillance instrument. — Retrieved from: ____________________

CASE 9 Vioxx Withdrawal 1999 – 2004

1. Bombardier, C. et al. (2000), “Comparison of upper gastrointestinal toxicity of rofecoxib and naproxen in patients with rheumatoid arthritis,” VIGOR trial, NEJM 343(21):1520–1528 — the five-fold myocardial-infarction signal and the naproxen-protective hypothesis. — Retrieved from: ____________________

2. Bresalier, R. et al. (2005), “Cardiovascular events associated with rofecoxib in a colorectal adenoma chemoprevention trial,” APPROVe, NEJM 352(11):1092–1102 — placebo-controlled confirmation of cardiovascular risk; early trial termination. — Retrieved from: ____________________

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6. FDA Amendments Act of 2007 (P.L. 110-85) and FDA Sentinel Initiative documentation (2008–present) — REMS authority, post-market study requirements, and active distributed-data post-market surveillance. — Retrieved from: ____________________

CASE 11 California Nurse Staffing Ratios — Legislating a Capability Requirement 2004 – 2010

1. Aiken, Sloane, Cimiotti, Clarke, Flynn, Seago, Spetz, & Smith (2010), “Implications of the California Nurse Staffing Mandate for Other States,” Health Services Research 45(4):904–921, doi:10.1111/j.1475-6773.2010.01114.x. — Retrieved from: ____________________

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4. Spetz, Chapman, Herrera, Kaiser, Seago, & Dower (2009), “Assessing the impact of California’s nurse staffing ratios on hospitals and patient care,” California HealthCare Foundation — implementation-period analysis. — Retrieved from: ____________________

CASE 12 ACGME 80-Hour Resident Duty-Hour Reform 2003–2017

1. B. H. Lerner, The Libby Zion Case and the Reform of Medical Education (2006); and the 1989 New York State (Bell Commission) duty-hour regulations — the origin of the reform. — Retrieved from: ____________________

2. Accreditation Council for Graduate Medical Education, Common Program Requirements (2003 and 2011 revisions) — the 80-hour weekly cap and the 16-hour first-year shift limit. — Retrieved from: ____________________

3. D. A. Asch, K. Y. Bilimoria & S. V. Desai, "Resident Duty Hours and Medical Education Policy — Raising the Evidence Bar," NEJM 376: 1704–1706 (2017); Institute of Medicine (Ulmer, Wolman & Johns, eds.), Resident Duty Hours: Enhancing Sleep, Supervision, and Safety (2008) — hand-offs and continuity trade-offs. — Retrieved from: ____________________

4. Bilimoria, Chung, Hedges et al., "National Cluster-Randomized Trial of Duty-Hour Flexibility in Surgical Training" (FIRST), NEJM 374: 713–727 (2016). doi:10.1056/NEJMoa1515724. — Retrieved from: ____________________

5. Silber, Bellini, Shea et al., "Patient Safety Outcomes under Flexible and Standard Resident Duty-Hour Rules" (iCOMPARE), NEJM 380: 905–914 (2019). doi:10.1056/NEJMoa1810642. — Retrieved from: ____________________

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CASE 13 Implementation Science in Healthcare — The 17-Year Gap

ongoing

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2008 – present

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1986 – present

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CASE 29 Bar-Code Medication Administration — Cue at the Point of Care

2010

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CASE 30 Surgical Skill Video Peer-Rating Predicts Complications 2013

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CASE 31 Language of Surgery / JIGSAWS — Decomposing Skill into Measurable Units 2006 – 2016

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CASE 32 Annual-Screening UI Redesign + CDS at University of Missouri Health Care 2020

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CASE 33 Alert-Fatigue Redesign — Cutting EHR Alerts Without Cutting the Safety Signal 2019 – 2024

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CASE 34 COMPOSER Sepsis Prediction — The Third Clinical-AI Sepsis Case 2022 – 2024

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